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Arthroscopic anterior cruciate ligament surgery: Results of autogenous patellar tendon graft versus the Leeds-Keio synthetic graft Five year follow-up of a prospective randomised controlled trial

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A B S T R A C T

We conducted a prospective, randomised controlled trial comparing anterior cruciate ligament reconstruction using middle third patellar tendon graft (PT) to synthetic Leeds-Keio (LK) ligament. The patients were randomised (26 PT, 24 LK). Subjective knee function was classified (Lysholm, Tegner activity, IKDC scores), laxity was measured (Lachman test, Stryker laxometer), and functional ability was assessed (one-hop test). There were no significant differences between Lysholm or IKDC scores at any stage by 5 years. Significant differences were found between the groups at 2 years for Tegner activity scores, laxity and one-hop testing. By 5 years there were no significant differences. Clinical equivalence was demonstrated between the two groups for the Lysholm score and one-hop test but not for the Tegner activity score at 5 years.
The use of the LK ligament has been largely abandoned due to reports of its insufficiency. Our results demonstrate that it is not as inferior as one might expect. We conclude that the results of LK ligament ACL reconstruction are as acceptable as those using PT. It may provide an additional means of reconstruction where no suitable alternative is present.

1. Introduction

Autogenous patellar tendon graft (PT) has been used for many years to reconstruct the anterior cruciate ligament (ACL) of the knee [1,2]. Autogenous tissue has been shown to be weaker than normal ACL tissue post-operatively [3], weaker than synthetic material and demonstrate considerable variability in healing [4]. Much research has therefore been undertaken into the use of synthetic alternatives to reconstruct the ACL [5].

The Leeds-Keio (LK) artificial ligament is a woven polyester synthetic graft, which is designed to provide a scaffold for in-growth of a neo-ligament. There have been several papers published since our trial began, suggesting that the LK ligament is inferior to patellar tendon graft [6–9]. However, none of these papers are of the prospective randomised conrolled type. One paper written just prior to our trial commencing is a randomised prospective trial comparing PT with LK for ACL reconstruction [10]. This study suggested that the LK ligament was inferior to PT, based on the development of laxity. More recently, the results of a randomised, prospective trial comparing the Ligament Advancement Reinforcement System (LARS) with PT have been published with encouraging results for the synthetic ligament [11].

With such conflicting results in the literature, we wished to clarify whether the results of ACL reconstruction using the Leeds-Keio ligament were symptomatically, functionally and clinically acceptable in comparison to the patellar tendon graft.

2. Materials and methods

2.1. Eligibility criteria

A protocol was written and peer reviewed. The study received ethical approval from the Local Research Ethics Committee. Patients with arthroscopically proven unilateral ACL ruptures were recruited to take part in the study. Exclusion criteria were: previous or associated significant lower limb injury (including previous ACL reconstruction on either side, other injury to either limb which would leave residual deficit or prevent participation in the rehabilitation programme) and other lower limb disease (infection of bone or joint, metabolic bone disease, severe osteo or rheumatoid arthritis, neuro-vascular disease). Informed consent was obtained from all participants.

2.2. Settings and location

The patients were recruited prospectively from the senior author's (RBS) clinic by the clinical research fellow. The senior author, who is

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experienced with both procedures, carried out all operations between January 1995 and April 1999 using standardised techniques.

2.3. Randomisation

Patients were allocated to their treatment using a closed envelope system opened at anaesthetic induction. The randomisation was generated in Excel v4.0 (Microsoft Corporation) using three blocked random number tables with twenty patients in each. Patients were unaware of their treatment group allocation. No patient had their treatment changed from that allocated.

2.4. Details of interventions

For the PT group the middle third patellar tendon graft was harvested using the Neoligaments Graftologer (Xiros plc formerly Neoligaments Ltd, Leeds, UK). The LK group underwent ACL reconstruction with the synthetic Leeds-Keio ligament (Xiros plc formerly Neoligaments Ltd, Leeds, UK). All patients were placed in a hinged knee brace post-operatively and a standard rehabilitation protocol was used for each patient.

2.5. Outcome assessment

Patients were assessed pre-operatively and post-operatively at 6 months, as well as at 1, 2 and 5 years. The following data were collected: demographic details; Lysholm II score (primary outcome); Tegner activity level [12]; IKDC score; and clinical examination details including Lachman test, arthrometric testing using the Stryker laxometer (Stryker Corporation, Kalamazoo, Michigan, USA) [13] and the one-hop test [14]. Assessments were made by appropriately trained medical staff. It was not possible to blind the assessors from the patient's group allocation without adversely affecting the knee assessments. The patient's post-operative subjective assessment of knee function, harvest site pathology, post-ACL reconstruction arthroscopy details and post-reconstruction adverse events were also recorded.

2.6. Statistics

A power calculation to demonstrate clinical equivalence between the two groups using the Lysholm score was performed. For 80% power, using an equivalence margin of ± 12 for the Lysholm score, 30 patients were required in each group (assuming a common standard deviation of 15, an assumption of zero expected mean difference and a two-sided 5% significance level). Equivalence margins were chosen to reflect a difference in outcome which was felt would be clinically important. Results were analysed on an intention to treat basis. Analyses of covariance were performed for Lysholm score, Tegner activity levels, laxometer and one-hop testing. Tests for trend were carried out on the results for Lachman testing, subjective function and IKDC scoring. SPSS version 16.0 was used to analyse the data.

3. Results

3.1. Recruitment

Fifty patients (mean age 31.3 years SD 6.7 years) with arthroscopically proven unilateral ACL ruptures were recruited to take part in the study. Twenty-six patients (52%) underwent ACL reconstruction with middle third PT graft. Twenty-four patients (48%) underwent ACL reconstruction with the synthetic Leeds-Keio ligament. Initially it was intended to recruit 60 patients to the trial but, due to emerging literature casting doubt on the efficacy of synthetic ligaments the study was halted after the inclusion of 50 patients (Fig. 1).

3.2. Baseline data

Table 1 shows the demographic details of the patients assigned to each group. As would be expected with randomisation of patients in the study, there was no significant difference in the ages ($p = 0.65$) or injured sides ($p = 0.79$) between the two groups. The mean time from injury to operation was longer in the patients who were randomised to have a LK reconstruction, though this was not found to be significant ($p = 0.07$).

3.3. Arthroscopy findings at ACL reconstruction

The degree of degenerative change at reconstruction was recorded in detail for 38 cases. The majority of changes were Grade I or II and the frequency for each compartment was; patello-femoral compartment 6 (16%); medial compartment 19 (50%); and lateral compartment 14 (37%). The most significant changes were found in the medial compartment with 4 (11%) patients having Grade III or IV changes. There was no significant difference between the two groups with respect to these findings. Again given the randomisation of patients in the study no significant difference would be expected. Meniscal pathology was recorded in detail for 44 cases. There were tears present or previously resected in 25 (57%) medial menisci and 15 (34%) lateral menisci.

3.4. Outcomes at 5 years

At 5 years details for all patients were identified from the case notes. All patients were contacted and 46 were available for full clinical review, with the remainder providing details of post-operative interventions (but not clinical scores) over the telephone.

3.4.1. Lysholm score

The Lysholm scores for the two groups pre-operatively and at 6 months, 1 year, 2 years and 5 years are shown, Table 2. The Lysholm scores were significantly less for the LK group pre-operatively. The Lysholm scores for the two groups were not significantly different at any interval post-operatively. By 5 years the difference between the mean values for each group was 4.4 (95% confidence interval (CI) −2.8, 11.7). At 5 years clinical equivalence was demonstrated between the two groups using an equivalence margin of ± 12.

3.4.2. Tegner activity level

The Tegner activity levels for the two groups pre- and post-operatively are shown, Table 3. As with the Lysholm scores, the Tegner activity levels were significantly less in the LK group pre-operatively. The Tegner activity levels were significantly less in the LK group at 2 years post-operatively but were no different by 5 years. By 5 years the difference between the mean values for each group was 0.6 (95% CI −0.3, 1.6). At 5 years clinical equivalence was not demonstrated between the two groups using an equivalence margin of ± 1.5.

3.4.3. IKDC score

The overall IKDC scores for the two groups pre- and post-operatively are shown, Table 4. The p-value between the groups was not below 0.05 at any stage.

3.4.4. Lachman test

On objective manual clinical examination assessed with the Lachman test part of the IKDC score, the LK ligament was more lax post-operatively at 6 months and 2 years, Table 5. There was no difference between anterior draw and pivot shift for the two groups at any time post-operatively.

3.4.5. Arthrometric testing

The anterior translation measured with the Stryker laxometer for the two groups is shown, Table 6. There was no significant difference pre-operatively. At 1 year there was a trend towards greater laxity in the LK group, and at 2 years the laxity was significantly greater. At 5 years the laxity in both groups was the same.

3.4.6. One-hop test

The one-hop test for the two groups at the various time intervals is shown in Table 7. The one-hop test for the patients in the LK group was significantly less at 2 years. By 5 years there was no significant difference. By 5 years the difference between the mean values for each group was 0.6 (95% CI −6.6, 7.8). At 5 years clinical equivalence was demonstrated between the two groups using an equivalence margin of ± 12.

3.4.7. Post operative subjective assessment of knee function

The patients' post-operative subjective assessment of their knee function is shown, Table 8. There was no significant difference at any point post-operatively.

3.4.8. Harvest site pathology

Harvest site pathology was assessed by anterior knee pain scores and the ability to kneel, as well as documenting anterior knee tenderness on examination. Between the two groups there was: significantly greater anterior knee pain in the PT group at 1 year ($p < 0.01$); no significant difference in ability to kneel at any stage; and significant tenderness in the PT group at 6 months ($p = 0.02$) which improved at 1 year ($p = 0.08$). All other comparisons were not significantly different for these factors.

3.4.9. Post-ACL reconstruction arthroscopy

Six of the PT group and ten of the LK group required one or more arthroscopies post-reconstruction, Table 9. In each group the majority of the grafts were satisfactory,

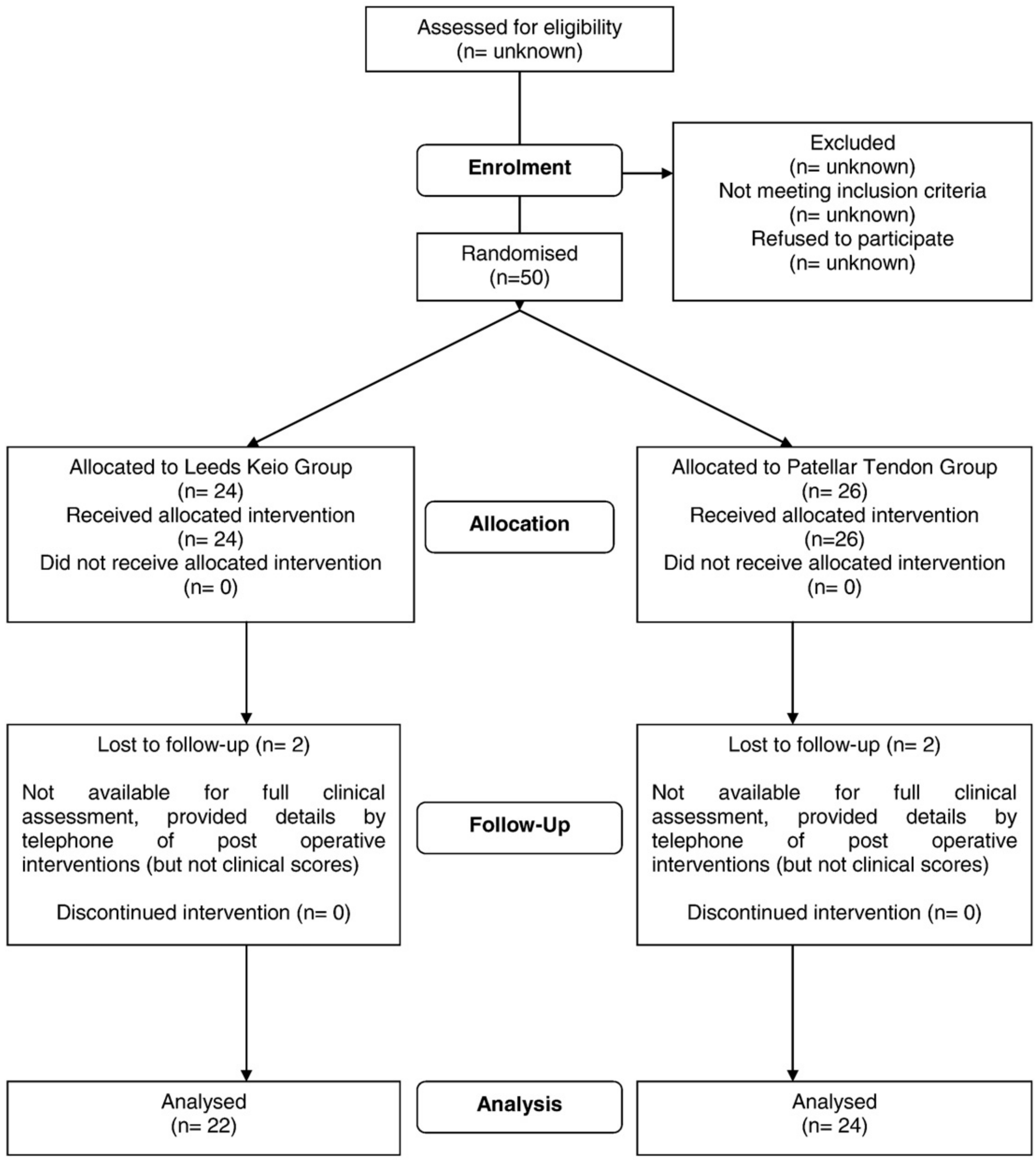

Fig. 1. Study flow chart.

although partial disruption or fraying of the graft was recorded in three of the LK group and two of the PT group, with a further two of the PT group having completely ruptured grafts.

3.4.10. Post-reconstruction adverse events

There were 11 further injuries documented during work or sporting activities (four LK, seven PT). At the five year stage: one patient in the LK group had undergone revision to a PT graft (following injury); and one patient in the PT group had undergone revision to a LK graft (also following injury), which had subsequently ruptured following a further injury requiring re-revision with an allograft.

Table 1
Demographic details.

	LK Group (95% CI)	PT Group (95% CI)
Number of patients	24	26
Sex (M:F)	21:3	19:7
Mean age (yrs)	31.7 (29.0–34.5)	30.9 (28.1–33.6)
Injured knee (L:R)	13:11	14:12
Mean time from injury to operation (months)	55 (35–75)	33 (19–47)

4. Discussion

In this study, the LK ligament was no more lax at 5 years and using the Lysholm and IKDC scores there was no statistically significant difference, although as in a paper by Engström et al. [10] there was a tendency towards a higher IKDC grade for the PT group. At 5 years clinical equivalence was demonstrated between the results for the

Table 2
Lysholm scores for the two groups pre- and post-operatively.

	Mean values		Adjusted mean values #		*p*-value #
	LK	PT	LK	PT	
Pre-op	62	69			
6 months	87	84	88 (82–93)	84 (78–89)	0.30
1 year	90	89	91 (87–95)	88 (85–92)	0.30
2 years	89	88	91 (85–96)	86 (81–92)	0.34
5 years	87	92	88 (83–93)	90 (85–95)	0.59

from analysis of covariance model (covariate = pre-op).

Table 3
Tegner scores for the two groups pre- and post-operatively.

	Mean values		Adjusted mean values #		*p*-value #
	LK	PT	LK	PT	
Pre-op	3.0	3.8			
6 months	4.6	4.1	4.6 (4.1–5.0)	4.1(3.6–4.5	0.14
1 year	5.1	5.2	5.1 (4.5–5.7)	5.2 (4.7–5.7)	0.83
2 years	5.2	6.2	5.2 (4.6–5.8)	6.2 (5.6–6.8)	0.028
5 years	5.5	6.2	5.4 (4.7–6.1)	6.2 (5.6–6.9)	0.10

from analysis of covariance model (covariate = pre-op).

two groups using the Lysholm score and one-hop test though not for the Tegner score.

There was a difference between both groups with respect to increased harvest site pathology in the PT group early on. This was also true for Tegner scores and laxity/function testing in the LK group at 2 years, which was not significant at 5 years. Looking more closely at these findings, it is seen that the activity scores and function tests gradually improve in both groups. In the PT group this reached a plateau at 2 years, whereas this was 5 years in the LK group. The slower recovery in the LK group explains the difference at 2 years for these factors. The difference in laxity between the two groups at 2 years is more difficult to explain. However, as the mean difference between the two groups is only 1.1 mm, this may simply be a reflection of the inaccuracy of the laxometer which we have previously reported [25]. These findings concur with the study of Engström et al. [10].

Initial arthroscopic assessment of the LK ligament was promising [15], showing 'ligamentisation' of the reconstructed ligament by 18 to 24 months. A study of 25 patients using the LK ligament in our unit also gave encouraging results [16], with 20 out of 25 LK ligaments at almost 5 years post-reconstruction being classified as 'good' or 'excellent' and over 50% returning to their pre-accident level of activity. Subsequent studies using arthroscopic and histological examination of the reconstructed ligament have suggested that the LK ligament was not particularly useful as a scaffold-type of artificial ligament [17,18], showing a lack of collagenisation and in-growth. Other studies suggested that while patients were less disabled than pre-operatively, objective results showed the LK ligament to be somewhat inferior to normal ACL tissue [19,20]. Histological examination of failed LK ligaments has shown that a chronic inflammatory response inhibits production of the neo-ligament [21,22]. However, to date there is no literature regarding the histology of an intact LK ligament. More recently, two studies from Japan have once again showed favourable results using the LK ligament [23,24], especially with a minor modification to the technique [24].

In considering our results, it is interesting that despite randomisation, the pre-operative Lysholm and Tegner scores for the LK group in our study are significantly less than the PT group. This may be a reflection of the difference in time from injury to operation between the groups. We had initially hoped to recruit 60 patients for this trial ensuring matched groups, but we felt compelled to stop recruiting because of the emerging literature suggesting that the LK ligament was unsatisfactory. Further statistical advice following curtailment of recruitment into the study was sought. We have been advised that our results have not been invalidated by the decision to stop recruitment because this decision was made based on the emerging literature. Despite the pre-operative disparity we have shown that the results are equivalent between the two groups for the Lysholm score and one-hop test. A potential limitation of our study is the increased possibility of spurious significant findings due to multiple testing of patients at several time points. Bearing our results in mind and in hindsight it would have been preferable to continue the trial. Although unable to reverse our decision, we would recommend other researchers to carefully consider their options if faced with a similar situation.

The pre-operative clinical examination was performed in the pre-operative assessment clinic and not at the time of surgery under anaesthetic. This allowed direct comparison with the post-operative clinical examination scores. As a result two patients in the PT group are recorded as having negative Lachman tests, both of whom at EUA had positive Lachman and pivot shift tests. This suggests that the pre-operative examination was modified by hamstrings activity guarding against testing. All patients in the trial had arthroscopically proven ACL tears, which correlated with the clinical findings.

The mean age at reconstruction is higher than might be expected compared to studies commenced currently, which was influenced by a number of the patients presenting with initially undiagnosed and

Table 4
IKDC scores for the two groups pre- and post-operatively.

	Pre-op		6 months		1 year		2 years		5 years	
	n (%)		*n* (%)		*n* (%)		*n* (%)		*n* (%)	
IKDC score	LK	PT	LK	PT	LK	PT	LK	PT	LK	PT
A	0 (0)	0 (0)	0 (0)	1 (5)	2 (11)	2 (9)	1 (5)	2 (9)	2 (9)	7 (29)
B	0 (0)	0 (0)	6 (29)	5 (26)	7 (39)	13 (56)	10 (48)	14 (64)	11 (50)	12 (50)
C	7 (30)	11 (42)	8 (38)	7 (37)	8 (44)	5 (22)	6 (28)	5 (23)	7 (32)	4 (17)
D	16 (70)	15 (58)	7 (33)	6 (32)	1 (6)	3 (13)	4 (19)	1 (4)	2 (9)	1 (4)
p-value			0.71		0.83		0.11		0.06	

Table 5
Lachman test for the two groups pre- and post-operatively.

	Pre-op		6 months		1 year		2 years		5 years	
	n (%)		*n* (%)		*n* (%)		*n* (%)		*n* (%)	
Lachman test	LK	PT	LK	PT	LK	PT	LK	PT	LK	PT
0–2 mm	0 (0)	2 (8)	10 (48)	16 (76)	10 (56)	14 (61)	9 (43)	16 (73)	15 (68)	16 (70)
3–5 mm	12 (50)	16 (61)	8 (38)	5 (24)	8 (44)	9 (39)	12 (57)	6 (27)	7 (32)	7 (30)
6–10 mm	10 (42)	6 (23)	3 (14)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)
>10 mm	2 (8)	2 (8)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)
p-value			0.028		0.73		0.05		0.92	

Table 6
Side to side difference in anterior translation at 30 degrees of flexion for the two groups pre- and post-operatively (mm).

	Mean values		Adjusted mean values #		*p*-value #
	LK	PT	LK	PT	
Pre-op	5.0	3.9			
6 months	1.2	1.0	1.2 (0.3–2.1)	1.0 (0.1–2.0)	0.82
1 year	1.8	0.8	2.1 (1.1–3.2)	0.7 (−0.2–1.6)	0.043
2 years	2.0	0.9	2.2 (1.4–3.0)	0.6 (−0.2–1.4)	0.009
5 years	0.6	0.8	0.6 (−0.3–1.5)	0.7 (−0.2–1.6)	0.88

from analysis of covariance model (covariate = pre-op).

Table 7
One-hop test for the two groups pre- and post-operatively (percentage ratio of injured to non injured leg).

	Mean values		Adjusted mean values #		*p*-value #
	LK	PT	LK	PT	
Pre-op	71	72			
6 months	68	74	90 (62–79)	73 (64–82)	0.69
1 year	82	89	78 (68–89)	91 (82–100)	0.078
2 years	82	93	81 (73–89)	95 (88–102)	0.012
5 years	91	91	88 (83–94)	92 (87–97)	0.33

from analysis of covariance model (covariate = pre-op).

subsequently chronic injuries. This is confirmed by the high mean time from injury to surgery, and reflects the degree of degenerative changes and meniscal damage found at reconstruction. This will certainly have had a bearing on the high number of patients requiring subsequent arthroscopic surgery, although some relate to patients returning to their previous level of activity and sustaining a new injury.

The only other randomised, prospective trial concerning the LK and PT are the intermediate results (follow-up varying from 12 to 50 months) published by Engström et al. [10]. Their surgery was performed by three different surgeons. They also stated that the uninjured knee was to be used as a control, but in three of their patients both knees were injured. They demonstrated that pivot shift and anterior laxity were significantly greater in the LK group and from this it was concluded that the LK ligament did not fulfil the requirements for a satisfactory result in ACL reconstruction with regard to knee joint stability. Regarding function however, they showed no difference between the two groups for either the Lysholm score or IKDC level, despite a tendency to a higher IKDC grade for the PT group.

We feel that symptoms and function are more important outcome measurements following ACL reconstruction than laxity measurement. Function and symptoms have been shown to correlate well with patient satisfaction [26], while examination findings correlate less well with the patient-reported scores [27,28]. Hyder et al. also questioned whether objective arthrometric measurements were appropriate or meaningful, finding that they did not correlate significantly with clinical outcome [29]. It has been reported recently in a trial of PT versus two-strand hamstring for ACL reconstruction that although from the perspective of clinical assessment alone the PT was superior, the two techniques produced similar outcomes with regard to patient satisfaction, activity level and knee function [30].

The reason for these findings is debatable. For the majority of patients ACL reconstruction does improve knee stability directly. However, for those ligaments that perform less well on clinical examination some other factors must be at work. It is possible that the functional outcome of ACL reconstruction is dependent on proprioceptive ability rather than laxity of the ligament, which may recover in reconstructed patients and may remain even if the graft subsequently becomes non-functional. Indeed it has been shown that objective measurements of proprioception correlate well with both function and patient satisfaction in the patient who has undergone ACL reconstruction [31,32].

Currently with the ACL being reconstructed increasingly with hamstring tendons with good results [33], the usefulness of synthetic ligaments must be questioned. The advantage that a synthetic ligament has over autologous tissue is that it can be used for revision surgery if other options are not available, because of previous injuries or surgical procedures, and it avoids the need for allograft. In addition, it can be used in combination with other autogenous tissue, where complex ligament reconstructions are being undertaken. Data from Canada [11,28] using the synthetic Ligament Advanced Reinforcement System (LARS) have shown encouraging results.

Table 9
Indications for post-reconstruction arthroscopy.

	Reconstruction Type	
	LK	PT
No. of patients	10	6
No. of procedures	10	9
Symptoms		
Pain	7	5
Instability	1	3
Reduced ROM	1	2
Locking	2	1
Swelling	2	1
Main findings		
Medial meniscal tear	5	2
Graft damage	3	4
Early osteoarthritis	4	2
Adhesions	2	2
Loose body	1	0

5. Conclusion

In summary, based on functional outcome and patient satisfaction, a comparable outcome in ACL reconstruction using the Leeds-Keio ligament or autogenous patellar tendon graft has been demonstrated in this study. Clinical equivalence for the results at 5 years between Leeds-Keio ligament and patellar tendon graft has been demonstrated as assessed with the Lysholm score and one-hop test. As such despite the limitations of the study, including the fact that it is a single

Table 8
Patient's subjective function post-operatively.

	6 months		1 year		2 years		5 years	
	n (%)		*n* (%)		*n* (%)		*n* (%)	
	LK	PT	LK	PT	LK	PT	LK	PT
Normal	1 (5)	2 (10)	0 (0)	3 (14)	4 (18)	9 (39)	3 (14)	6 (25)
Improved	13 (62)	15 (71)	17 (94)	18 (79)	15 (68)	13 (57)	17 (77)	16 (67)
No change	3 (14)	3 (14)	1 (6)	1 (4)	3 (14)	1 (4)	2 (9)	2 (8)
Worse	4 (19)	1 (5)	0 (0)	1 (4)	0 (0)	0 (0)	0 (0)	0 (0)
p-value	0.17		0.71		0.085		0.44	

surgeon series, we feel that, if used appropriately, the Leeds-Keio ligament may still have a role in ACL reconstructive surgery in situations where there are no suitable alternatives.

6. Conflict of interest

None.

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